KARNATAKA SCHOOL EXAMINATION AND ASSESSMENT BOARD II PUC EXAM-2 APRIL-MAY 2025

Subject: 35-Mathematics SCHEME OF VALUATION MAX. MARKS: 80

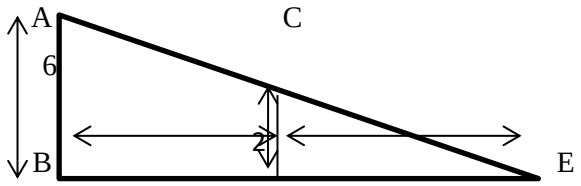
Instructions:

a) Any answer by alternate method should be valued and suitably awarded.

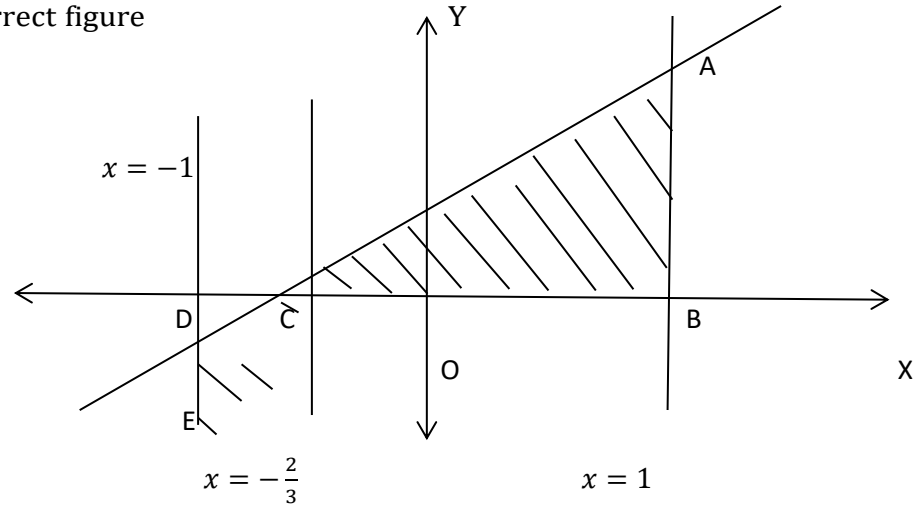
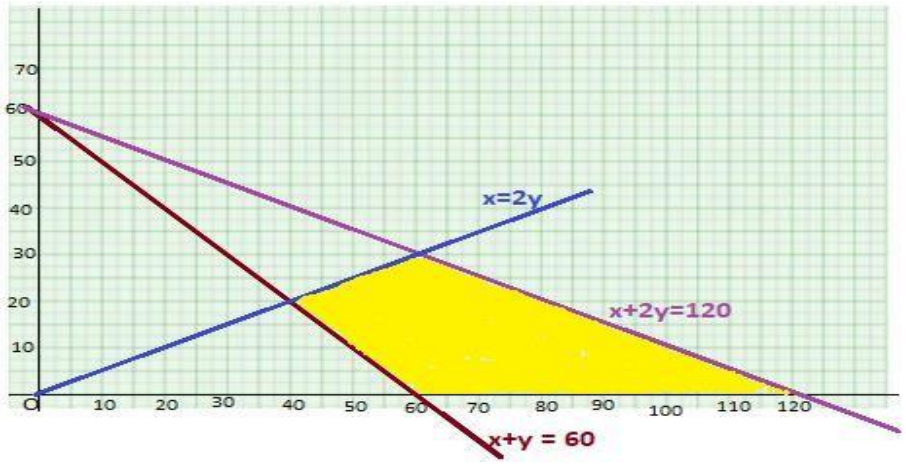
b) All answers (including extra, struck off and repeated) should be valued. Answers with maximum marks must be considered.

Q No:	PART A	Marks
I. 1	c) or writing Transitive	1
2	b) or writing $(0, \pi)$	1
3	c) or writing Statement 1 is true but statement 2 is false	1
4	a) or writing 4	1
5	b) or writing 64	1
6	d) or writing $\frac{1}{x\,logx}$	1
7	c) or writing 0	1
8	d) or writing 126	1
9	a) or writing $\frac{e^x}{x} + C$	1
10	a) or writing $\frac{\pi}{12}$	1
11	b) or writing $4\vec{b} - \vec{a}$	1
12	c) or writing $\pm\frac{1}{\sqrt{3}}$	1
13	b) or writing $0, 0, -1$	1
14	a) or writing P(A) = P(B)	1
15	d) or writing 0.12	1
II . 16	0	1
17	-1	1
18	3	1
19	-2	1
20	1	1
	PART B	
21	Writing $\begin{vmatrix} 1 & 1 & 1 \\ x & 1 & 0 \\ y & 3 & 0 \end{vmatrix} = 0$ **OR** $\begin{vmatrix} 1 & x & y \\ 1 & 1 & 3 \\ 1 & 0 & 0 \end{vmatrix} = 0$	1
	Getting $3x - y = 0$ **OR** $3x = y$ **OR** $y - 3x = 0$	1
22	Getting $\frac{dy}{dx} + cosy\ \frac{dy}{dx} = -sinx$	1

	Writing $\frac{dy}{dx} = \frac{-sinx}{1+cosy}$	1
23	Writing $f'(x) = 3x^2$	1
	Getting absolute maximum value of f(x) in $[-2,2] = 8$	1
24	Writing $f'(x) = 2x - 4$ **OR** $x = 2$	1
	Getting $x \in (2, \infty)$	1
25	Writing $\int \frac{x^3-x^2+x-1}{x-1}\, dx = \int x^2 + 1\, dx$	1
	Getting $\int \frac{x^3-x^2+x-1}{x-1}\, dx = \frac{x^3}{3} + x + C$	1
26	Writing $-\frac{1}{y^2}\, dy = 4x\, dx$	1
	Getting $\frac{1}{y} = 2x^2 + C$ **OR** $y = \frac{1}{2x^2+C}$	1
27	Getting $\vec{a} \times \vec{b} = 5\hat{\imath} + \hat{\jmath} - 4\,\hat{k}$	1
	Getting area of triangle $\frac{\sqrt{42}}{2}$ sq. units	1
28	Writing $cos\theta = \frac{\vec{b_1}\cdot\vec{b_2}}{\lvert\vec{b_1}\rvert\lvert\vec{b_2}\rvert}$ **OR** $cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1{}^2+b_1{}^2+c_1{}^2}\sqrt{a_2{}^2+b_2{}^2+c_2{}^2}}$	1
	Getting $\theta = cos^{-1}\frac{2}{3}$	1
29	Writing P(black ball in first draw) = P(E) $= \frac{26}{52} = \frac{1}{2}$ **OR** P(black ball in second draw) = P(F|E) $= \frac{25}{51}$	1
	Getting $P(E \cap F) = P(E) \times P(F\|E) = \frac{1}{2} \times \frac{25}{51} = \frac{25}{102}$	1
	PART C	
30	Reflexive: $\forall a \in Z, 2\ divides\ (a-a) = 0 \Rightarrow (a,a) \in R$	1
	Symmetric: $let\ (a,b) \in R \Rightarrow 2\ divides\ (a-b) \Rightarrow 2\ divides\ (b-a) \Rightarrow (b,a) \in R$	1
	Transitive: $let\ (a,b)$ & $(b,c) \in R \Rightarrow 2\ divides\ (a-b)$ & $(b-c)$ $\Rightarrow 2\ divides\ (a-b) + (b-c) \Rightarrow 2\ divides\ (a-c) \Rightarrow (a,c) \in R$ and writing conclusion.	1
31	Writing cosA = $\frac{4}{5}$, sinA = $\frac{3}{5}$, cosB = $\frac{12}{13}$ and sinB = $\frac{5}{13}$	1
	Getting cos(A−B) = $\frac{63}{65}$	1
	Writing $cos^{-1}\frac{4}{5} - cos^{-1}\frac{12}{13} = cos^{-1}\frac{63}{65}$ OR any other Alternate method allot appropriate marks	1
32	Writing $A = A^T\ and\ B = B^T$	1
	Getting: $(AB - BA)^T = BA - AB$	1

	Getting: $(AB-BA)^T=-(AB-BA)$ and writing conclusion	1
33	Getting $\frac{dy}{d\theta}=-a\ sin\theta$	1
	Getting $\frac{dx}{d\theta}=a\ (1-cos\theta)$	1
	Getting $\frac{dy}{dx}=\frac{-a\ sin\theta}{a\ (1-cos\theta)}$ **OR** $\frac{dy}{dx}=-\ cot\frac{\theta}{2}$	1
34	Writing figure 	1
	Writing $\frac{AB}{CD}=\frac{x+y}{y}$ and Getting $2y=x$	1
	Getting $\frac{5}{2}\ km/hr$	1
35	Writing $\frac{3x-2}{(x+1)^2(x+3)}=\frac{A}{(x+3)}+\frac{B}{(x+1)}+\frac{C}{(x+1)^2}$	1
	Getting: $A=-\frac{11}{4},\ B=\frac{11}{4}$ and $C=-\frac{5}{2}$	1
	Getting $\int\frac{3x-2}{(x+1)^2(x+3)}dx=\frac{11}{4}\ log\left\lvert\frac{x+1}{x+3}\right\rvert+\frac{5}{2(x+1)}+c$	1
36	Writing $\left\lvert\vec{a}+\vec{b}+\vec{c}\right\rvert^2=\lvert\vec{a}\rvert^2+\left\lvert\vec{b}\right\rvert^2+\lvert\vec{c}\rvert^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$	1
	Substituting the values $\lvert\vec{a}\rvert, \left\lvert\vec{b}\right\rvert, \lvert\vec{c}\rvert$ and $\left\lvert\vec{a}+\vec{b}+\vec{c}\right\rvert$	1
	Getting $\mu=-25$	1
37	Writing $\overrightarrow{a_2}-\overrightarrow{a_1}=\hat{\imath}-3\hat{\jmath}-2\hat{k}$	1
	Getting $\overrightarrow{b_1}\times\overrightarrow{b_2}=-3\hat{\imath}+3\hat{k}$ **OR** $\left\lvert\overrightarrow{b_1}\times\overrightarrow{b_2}\right\rvert=\sqrt{18}$	1
	Getting shortest distance $p=\frac{(\overrightarrow{a_2}-\overrightarrow{a_1})\cdot(\overrightarrow{b_1}\times\overrightarrow{b_2})}{\lvert\overrightarrow{b_1}\times\overrightarrow{b_2}\rvert}=\frac{3}{\sqrt{2}}$	1
38	Writing $P(E_1)=P(E_2)=P(E_3)=\frac{1}{3}$ **OR** $P(A/E_1)=1,\ P(A/E_2)=0,\ P(A/E_3)=\frac{1}{2}$	1
	Writing $P(E_1/A)=\frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)+P(E_3)P(A/E_3)}$	1
	Getting $P(E_1/A)=\frac{2}{3}$	1
	PART D	
39	Writing $x_1, x_2 \in R\ such\ that\ f(x_1)=f(x_2)\Rightarrow 3-4x_1=3-4x_2$	1
	Getting $x_1=x_2$, $\therefore$f is one-one.	1
	Writing y $\in$ B and let f(x) = y	1
	Getting $x=\frac{3-y}{4}\in$ R, $\therefore f$ is onto.	1

	Writing f is bijective function	1
40	Getting $A+B=\begin{bmatrix}0 & 7 & 8\\ -5 & 0 & 10\\ 8 & -6 & 0\end{bmatrix}$	1
	Getting $(A+B)C=\begin{bmatrix}10\\20\\28\end{bmatrix}$	1
	Getting $AC=\begin{bmatrix}9\\12\\30\end{bmatrix}$	1
	Getting $BC=\begin{bmatrix}1\\8\\-2\end{bmatrix}$	1
	Getting $AC+BC=\begin{bmatrix}10\\20\\28\end{bmatrix}$ and writing $(A+B)C=AC+BC$	1
41	Writing AX=B $\Rightarrow \begin{bmatrix}4 & 3 & 2\\ 2 & 4 & 6\\ 6 & 2 & 3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}60\\90\\70\end{bmatrix}$ **OR** A = $\begin{bmatrix}4 & 3 & 2\\ 2 & 4 & 6\\ 6 & 2 & 3\end{bmatrix}$, B = $\begin{bmatrix}60\\90\\70\end{bmatrix}$, X = $\begin{bmatrix}x\\y\\z\end{bmatrix}$ **OR** Getting $\lvert A\rvert = 50 \neq 0$ **Note:** Award a mark, if student writes directly $\lvert A\rvert = 50$	1
	Getting $adjA=\begin{bmatrix}0 & -5 & 10\\ 30 & 0 & -20\\ -20 & 10 & 10\end{bmatrix}$ **Note:** If any 4 cofactors are correct award 1 mark.	2
	Writing $X = A^{-1}B=\frac{1}{50}\begin{bmatrix}0 & -5 & 10\\ 30 & 0 & -20\\ -20 & 10 & 10\end{bmatrix}\begin{bmatrix}60\\90\\70\end{bmatrix}$	1
	Getting x = 5, y = 8, z = 8	1
42	Getting $\frac{dy}{dx}=\frac{2\tan^{-1}x}{(1+x^2)}$.	1
	Writing $(1+x^2)\frac{dy}{dx}=2\tan^{-1}x$	1
	Getting $(1+x^2)\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=\frac{2}{(1+x^2)}$	1
	Writing $(1+x^2)[(1+x^2)\frac{d^2y}{dx^2}+2x\frac{dy}{dx}]=2$	1
	Writing $(1+x^2)^2\frac{d^2y}{dx^2}+2x(1+x^2)\frac{dy}{dx}=2$	1
43	Writing $\frac{1}{(x^2-a^2)}=\frac{1}{2a}\left[\frac{1}{x-a}-\frac{1}{x+a}\right]$	1
	Writing $\int\frac{1}{(x^2-a^2)}dx=\frac{1}{2a}\left[\int\frac{1}{x-a}dx-\int\frac{1}{x+a}dx\right]$	1
	Getting $\int\frac{1}{(x^2-a^2)}dx=\frac{1}{2a}log\left\lvert\frac{x-a}{x+a}\right\rvert+C$	1
	Writing $\int\frac{1}{(4x^2-9)}dx=\int\frac{1}{(2x)^2-3^2}dx$ **OR** $\frac{1}{4}\int\frac{1}{x^2-\left(\frac{3}{2}\right)^2}dx$	1

	Getting $\int \frac{1}{(4x^2-9)}dx = \frac{1}{12}log\left\|\frac{2x-3}{2x+3}\right\| + C$	1
44	Writing correct figure 	1
	Writing Area = Area of region ABCA + Area of region DCED	1
	Getting Area of region ABCA $=\int_{-\frac{2}{3}}^{1}(3x+2)dx = \frac{25}{6}$	1
	Getting Area of region DCED $=\left\|\int_{-1}^{-\frac{2}{3}}(3x+2)dx\right\| = \frac{1}{6}$	1
	Getting Area $= \frac{13}{3}$ square units **Note:** Units are not compulsory	1
45	Writing $\frac{dy}{dx}+\frac{2}{x}y = x\,logx$ ***OR*** $P = \frac{2}{x},\ Q = x\,logx$	1
	Getting $IF = e^{\int P\,dx} = x^2$	1
	Writing $y(IF) = \int Q(IF)dx + C$ ***OR*** $yx^2 = \int x^3 logx\,dx + C$	1
	Writing $yx^2 = logx\ \frac{x^4}{4} - \int \frac{x^3}{4}dx + C$	1
	Getting $yx^2 = \frac{x^4 logx}{4} - \frac{x^4}{16} + C$ (Deduct 1 mark for not writing C)	1
	PART E	
46	 Drawing the graph (award 1 mark for correct two lines and 2 marks for correct three lines)	2

	Getting corner points $(60,0),(120,0),(60,30)$ *and* $(40,20)$ ***OR*** Showing corner points on the graph	1
	<table><tr><td>Corner points</td><td>Z=5x+10y</td></tr><tr><td>(60, 0)</td><td>300</td></tr><tr><td>(120, 0)</td><td>600</td></tr><tr><td>(40, 20)</td><td>400</td></tr><tr><td>(60, 30)</td><td>600</td></tr></table>	1
	Writing the minimum value of Z is 300 at $(60,0)$	1
	The maximum value of Z is 600 at all the points on the line segment joining $(60,30)$ *and* $(120,0)$	1
OR	Taking $x = a + b - t$ **OR** $a + b - x = t,\ dx = -dt$ $x = a \Rightarrow$ t = b and x = b $\Rightarrow$ t = a	1
	Getting $\int_a^b f(x)\,dx = -\int_b^a f(a+b-t)\,dt$	1
	Getting $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$	1
	Writing $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{5-x}}dx = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x}+\sqrt{x}}dx$	1
	Getting $2I = \int_2^3 1\,dx$	1
	Getting $I = \frac{1}{2}$	1
47	Getting $A^{-I} = \begin{bmatrix} 5 & -7 \\ -2 & 3 \end{bmatrix}$ and $B^{-I} = \frac{1}{2}\begin{bmatrix} 9 & -8 \\ -7 & 6 \end{bmatrix}$	1
	Getting $B^{-I}A^{-I} = \frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$	1
	Getting $AB = \begin{bmatrix} 67 & 87 \\ 47 & 61 \end{bmatrix}$	1
	Getting $(AB)^{-I} = \frac{1}{2}\begin{bmatrix} 61 & -87 \\ -47 & 67 \end{bmatrix}$ and writing $(AB)^{-I} = B^{-I}A^{-I}$	1
OR	Writing condition for continuity $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = f(2)$	1
	Writing $\lim_{x\to 2^-} f(x) = 4k$ ***OR*** $f(2) = 4k$	1
	Writing $\lim_{x\to 2^+} f(x) = 3$	1
	Getting $k = \frac{3}{4}$	1

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